

Challenges faced by Business Analysts eliciting User Requirements in the software industry: A survey-based study

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Abstract—Requirement gathering is considered as one of the most important activity in software development lifecycle (SDLC). If requirements are not gathered properly then there is huge expectation gap between what the customer desire and what the developer built. Currently, there are various techniques used to elicit software requirements such as interview, questionnaires, and prototyping etc. However, identify the common risks arise and adopting the best methodology in requirement gathering is still a topic of debate. In this paper, we performed a survey-based study featuring 21 participants aims to identify various aspects in requirement gathering process such as best approaches, techniques, problems occurred, risks involved, and benefits. The participants involved in this study are from diverse industrial roles (i.e., business analyst, software developer and technical lead etc.) and are working in requirement gathering process of real software projects. The main findings of this survey study shows that many participants (16/21, 76.2%) use interview approach followed by prototyping approach (13/21, 61.9%) to elicit the requirements. Moreover, related to potential risks, (9/21, 42.9%) participants responds insufficient user involvement and ambiguous requirements as most common risks experienced by business analyst, whereas (1/21, 4.8%) participant select gold plating as less frequent risk arise during requirement gathering.

Keywords – Requirement Gathering, Requirement Elicitation, Survey study

I. Introduction

Requirement engineering is collection of rules and procedures that guide on how requirements are gathered for a software system in well managed way [1]. In software terminology the foundation of a good and well-structured software depends upon requirements gathering¹ process, tools, and techniques like foundation of a building is based

upon its structure and baseline because the baseline is the core part of a building structure.

It is considered as most difficult task to (i) understand what customer want to develop and, (ii) then document those requirements. For this purpose, business analyst (or a person who gathers requirement) need a deep understanding of technical knowledge of the product, business domain knowledge, and different skills set such as interview, survey, prototype etc., which are required to elicit requirements from customer side [2]. There are various risks that can arise during requirement gathering process such as insufficient user involvement, ambiguous requirement, gold plating, and overlooked stakeholder etc. These risks can cause unclear and poorly written requirements, that ultimately lead to failure of software product. These unclear and poorly written requirements can consume more time and resources and increase the rework cost.

Many traditional techniques like Interviews, Questionnaire, Workshops, Domain Analysis, prototyping etc., are used in requirement elicitation process. These techniques are purely based on human effort, for example a Business analyst (BA) is primarily responsible for gathering requirements from customer and further analyze the whole system domain knowledge. The BA is a person who identified and write various requirements in software requirement specification document (SRS) such as Functional Requirements, non-functional Requirements and Ambiguous Requirements.

In this paper, we performed a survey-based study to identify various aspects of requirement gathering process that are experienced by business analyst or a person who

¹In the remainder of paper, the term requirement gathering, and requirement elicitation can be used interchangeable.

is responsible for requirement elicitation. Some of the aspects we explored in this paper are best methodology (technique), skills required, risks and benefits in requirement gathering activities. The survey questions considered in this study are partially inspired by a software engineering book [20]. To be specific, this paper makes the following key contributions:

- To best of our knowledge, this is first of its kind study aimed to identify different types of challenges encountered by business analyst during requirements elicitation.
- This study provides the overview of real practices, risks and benefits come across by the business analyst at the time of requirement gathering process.
- The replication package of this study is provided that includes the responses of various participants considered in this study. This package will serve as a basic step towards the future work for researchers and business analysts' community in requirement gathering domain.

The paper is organized as follows. Section II and III highlights the background and related work of the study. Section IV describes the design of the study, whereas section V provides the details of data extraction process used in this paper. Results of the study are reported in section VI and discussion is explained in section VII. Conclusion and future work are described in section VIII.

II. Background

There are numbers of techniques available that are helpful to manage software requirements. Some of common techniques used are discuss as the following.

Domain Background Knowledge: The background knowledge is the essential part of any system during requirement gathering. The reason to have background knowledge of a system is that it can easily identify the customer requirements and what our customer want to develop. However, this technique is time consuming as more time is required to gain background knowledge of any domain [2].

Interview: Interview is the useful technique in specific scenario when a customer or vender has core knowledge of their business domain and they are sure about the detail of information they needed from the customer. In this technique a business analyst and vendor or customer have face to face meeting or direct communication [3]. The only experience person who have required skill set will usually collects user requirements using interview approach.

Apprenticing: In Apprenticing technique requirements are collecting by observing users on their workplace. In this process the business analyst gather requirement from user environment. This approach is helpful when a

company have many employees and they have their different roles and responsibilities. This technique is time consuming activity, and a lot of hidden information can be missed using this technique [4].

Requirement Modeling: The types of requirement models are State diagram, Entity Relationship diagram, Sequence diagram and Use Case diagram. Using these requirement models requirement can be validate from user [5].

Brainstorming: Brainstorming is an innovative technique for gathering requirement with new ideas and solutions. This technique is use for specific domain like gamming apps, but this technique fails on customer business domain because a customer has better knowledge of his domain [3].

Workshops and forums: Workshops and forums technique are very helpful when you have required solutions of many problems or you have launched a new business with innovative brand, and you need to continue feedback then a workshop can be arranging with business experts, customer and vender sit together and start their discussions [4].

Prototyping: Prototyping technique is very helpful for gathering software requirements because a prototype can explain to customer that what the system he wants to develop and how its look like after development. When we are gathering requirements with other traditional approaches a lot of challenges and risk are associated with it [6]. Prototyping approach can present a clear-cut picture of user system what he wants to develop. But this approach is time consuming because in initial stage a first business experts gather user requirement and identify user domain requirements and then draw these requirements into a form of prototype [4].

III. Related Work

There are various studies conducted in the past which are focused-on suggestions, techniques, and methods for gathering software requirements [7]. *D. Carrizo et.al* [8] emphasized that the most important part to choose the right requirement technique at right time. The selection of right techniques depends upon the system environment and technical ability of Client. Whereas, *O. Emoghene et.al* [9] explained that the success of a software system depends upon various factors but one of the key factor is the selection of right requirement gathering technique at a right time. Some literature studies also focused on the performance aspect of software applications [13][14][15][16] and few studies are dedicate on software estimation [17][18][19].

Few studies in the past analyzed the different approaches used in requirement gathering process for ecample D. *Mishra et.al* [10] suggested that interview is not the best approach for gathering complex and modern system

requirements. In this situation we must be choose more than one requirement gathering techniques for better understanding. Further, L. Jiang *et.al* study findings that when we develop a complex system the requirements can be changes according to user need and it is not possible to gathering complex system requirements at initial level at beginnings. When we talk about requirement elicitation process for local client and as well as international client then we faced a lot of barriers (like language, time zone etc.). In this situation selection of right and proper elicitation technique by Business analyst is necessary for project success [11].

The study conducted by S. K *et.al* [12] showed that requirement gathering is the most critical part and during this process we can face many problems and challenges. One of the main problems and challenges is boundary of a customer requirements is not defined. When customer requirements boundary is not defined then we may be miss out necessary information which is the main part of the system. Sometime customer is unable to explain their requirements and sometimes customer feel uncomfortable and hesitate to explain their requirements. To overcome these problems and challenges, according to [12], when we gather user requirements, we should use the traditional approaches such as direct and Indirect approach. According to S. K *et.al* [12], the success of a project is not depends upon how many numbers of approaches are using for requirement gathering, instead the success of project is depending upon how you effectively and efficiently gather user requirements using these approaches according to different scenarios.

IV. Design of the Study

The *goal* of the study is to investigate current practices adopted in requirement gathering process with the *purpose* to understand common techniques, challenges and risks occur during requirement gathering process. The *context* of study consists of 21 participants which are currently working in the roles (such as business analyst, software engineer, and technical lead etc.) of requirement elicitation in various software companies. This survey study is conducted to address the answers to following research questions.

RQ₁ - What are the best approaches and practices used by Business Analyst during requirement gathering?

The selection of best requirement gathering approaches can depend on situation (i.e., depend on the types of software project going to develop or the types of users from which requirements must be gathered). Therefore, the purpose of *RQ₁* is to identify that what is the criteria used by business

analyst for the selection of the best requirement gathering approaches.

RQ₂ - How do Business Analysts identify different types of requirements (Functional, Non-Functional, Ambiguous)?

As we know that there are many types of requirements (functional, non-functional etc.) that are collected from user and documented in software requirement document (SRS). Therefore, the purpose of *RQ₂* is to address what are the criteria used by business analyst to identify the various type of requirements. For example, it is interesting to analyze if business analyst used any tool or manual approaches to identify various type of requirements.

RQ₃ - What are the common risks and challenges faced by Business Analysts during requirements gathering?

The goal of *RQ₃* is to identify what types of challenges, risks or difficulties experienced by business analyst during requirements elicitation. Moreover, this research question focused on how business analyst tackle with these challenges and risks.

To answer these research questions, we performed an online survey-based study from the experienced people (those who gather requirements in real software systems such as BA) from the Software Industry. As shown in Table I, we formulated the 25-survey questions (SQ) for our study. Out of these 25 questions, 16 survey questions are devised to answer the three main research questions of this paper. It can be shown in the Table I, shows the various survey questions (SQ) combines to answer a specific research question. The *NR* in the research question column indicates that the specific survey question is *not related* to any research question (RQ) of this study. Since this is a shorter version of a journal paper and we aimed to extend this work in future by performing a deeper analysis of the survey questions (SQ) which have *NR* fields. Furthermore, the replication package² of our study is publicly available featuring responses of participants and analyzed diagrams.

TABLE I: SURVEY QUESTIONS WITH THEIR CORRESPONDING RESEARCH QUESTION

ID	Survey Question	Research Question
SQ ₁	In your Company, Which person is responsible for gathering user requirements	NR
SQ ₂	How can you gather User Requirement?	NR
SQ ₃	If you are using any Automatic tool for Requirement Gathering, Please Enter tool Name.	NR

²Replication Package: <https://docs.google.com/spreadsheets/d/1MeAs-tFFwiQfKGQknUdELNjgcpUptXkTwbZB-Yv70PY/edit?usp=sharing>

SQ ₄	How can you Identify functional Requirement?	RQ ₂
SQ ₅	How can you Identify Non-functional Requirement?	RQ ₂
SQ ₆	How can you Identify Ambiguous Requirement?	RQ ₂
SQ ₇	Which type of Requirement is Important for User?	RQ ₁
SQ ₈	What type of method(s) do you use commonly while gathering user requirements?	RQ ₁
SQ ₉	How many numbers of meeting are required with client for gathering system Requirements	NR
SQ ₁₀	Can user easily explain that what type of system they need?	NR
SQ ₁₁	Is the User requirements are Stable or Changed With the passage of time?	NR
SQ ₁₂	Which type of Non-Functional requirement(s) consider as the most important part of the system?	RQ ₁
SQ ₁₃	What are the benefit(s) of High Quality Requirement Gathering Process?	RQ ₁
SQ ₁₄	Do you perform any risk analysis on requirements?	RQ ₃
SQ ₁₅	Which type of Risk factor(s) are involved while gathering the requirements?	RQ ₃
SQ ₁₆	Do you carry out a feasibility study before gathering a new project requirements?	RQ ₁
SQ ₁₇	Do you use any technique(s) to elicit user requirements?	RQ ₁
SQ ₁₈	Do you reuse requirements from other systems which have been developed in the same application area?	RQ ₁
SQ ₁₉	Do you define system boundaries?	RQ ₀
SQ ₂₀	What type of requirements you prioritize?	RQ ₁
SQ ₂₁	Do you focus on any guideline(s) while gathering user requirements?	RQ ₃
SQ ₂₂	What are some of the problem(s) you are faced when gathering requirements?	RQ ₃
SQ ₂₃	Which of the phase consider as the most critical?	NR
SQ ₂₄	What type of Skills Business Analyst must have to conduct or perform requirement gathering process?	NR
SQ ₂₅	What are the reasons for project failure?	RQ ₃

V. Data Extraction

We used standard google form to carry out online-based survey study. The study involves 21 participants perform various job responsibilities in industry as shown in

TABLE . Out of 21 participants, business analyst was large in number (i.e., 13/21), which includes both senior business analyst and only business analyst in our dataset followed by software engineer (5/21). The main reason to consider participants other than business analyst in our study is because there are many companies, which generally do not have a dedicated business analyst to elicit the requirements but instead any person from the team is performing business analyst job such as software engineer, quality assurance engineer, and technical lead etc.

For data analysis purpose, we moved the responses data into spreadsheet with respect to each survey question (SQ) of individual person. The idea is to have all the data at a one place in spreadsheet to easily analyze them. After the migration of these responses from online google form to spreadsheet then we analyzed the results with the help of different tables and diagrams.

TABLE II: SURVEY FORM FILLED BY DIFFERENT JOB RESPONSIBILITIES PEOPLE

Type of Job Responsibility	No of Participants
Senior Business Analyst	11
Senior Software Engineer	3
Business Analyst	2
Software Engineer	2
Business Development Director	1
Senior Quality Assurance Engineer	1
Technical Lead	1

VI. Results

This section is emphasized on the results obtained from the study. To answer the research questions, we designed an online survey form. A total of 21 number of responses received from various participants (Business Analyst, Senior Business Analyst, Senior Software Engineer, Technical Lead) involved in our study.

RQ₁ - What are the best approaches and practices used by Business Analyst during requirement gathering?

To answer this research question (*RQ₁*), we designed survey questions (*SQ₈, SQ₁₂, SQ₁₃, SQ₁₆, SQ₁₇, SQ₁₈, SQ₂₀*) as shown in Table I. Further, we analyzed the responses of these survey questions.

In *SQ₈* of the survey, we asked the participants about methods (approaches) commonly use during requirements gathering process. The results show that (16/21, 76.2%) participants adopt interview approach followed by prototyping approach (13/21, 61.9%) and questionnaire approach (8/21, 38.1%) as shown in the figure 1. This

makes sense because interview approach generally works efficiently when business analyst clearly knows what extent of information detail he wants to ask from the users. Therefore, business analyst practice interview approaches quite frequently in the requirement collection process. Prototype approach is another reliable way to elicit the requirements, in which business analyst show a prototype of software product to customer and collect the requirement. A few participants choose brainstorming approach (6/21, 28.6%) and workshop approach (1/21, 4.8%) during requirement gathering process. The selection of requirements gathering approaches purely depends upon type of software product going to build, situations, and environment., because we cannot select same approach for every software projects. But commonly approach used for requirement gathering that is interview approach as shown in figure 1. Hence, it will be interesting to observe that business analyst utilized many approaches during the requirement elicitation process. Further, when we asked about the requirement gathering technique (SQ_{17}), (14/21, 66.7%) participants supported scenario-based technique for gathering user requirements followed by use case technique (13/21, 61.9%).

Moreover, in SQ_{13} of the survey, we were curious to know from the participants about benefits of utilizing the best approaches for requirement gathering. The results indicate that (15/21, 71.4%) participants agreed on *all of these* option which means there are many benefits of practicing best approaches such as higher customer satisfaction, lower enhancement cost, fewer unnecessary and unused features, faster development and reduce development rework. While few participants (4/21, 19%) were supported reduce development cost and rework as key benefits. Finally, (3/21, 14.3%) participants choose higher customer satisfaction as shown in figure 2.

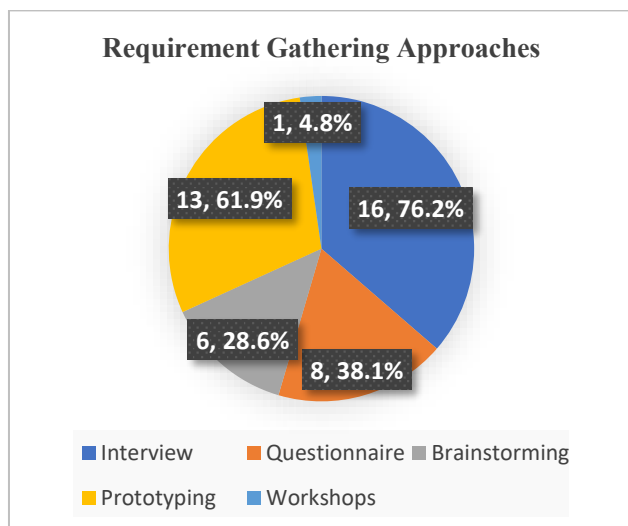


Figure. 1: Common approaches for requirement gathering

While asking about feasibility study from the participants (SQ_{16}), (17/21, 81%) respondents said they carry out feasibility study before start gathering the requirements of

new project. Whereas (2/21, 9.5%) of participants said that they do not perform any kind of feasibility study. The one of the reason why large numbers of participants perform feasibility study of a system because it plays very important role in requirement gathering process, because we will get to know background knowledge of the system that is going to be build. Hence, we will be able to understand user requirement in a better way and easily identify the need of users.

RQ₂ - Ho do Business Analysts identify different types of requirements (Functional, Non-Functional, Ambiguous)?

To answer this research question (RQ_2), we formulated survey questions (SQ_4 , SQ_5 , SQ_6) as shown in Table I. Further, we inspect the responses of these survey questions.

The SQ_4 is dedicated to knowing from the participants about how they detect functional requirements. Functional requirements are essential part of the system and play a vital role in the success of any software product. The results show all of (21/21, 100%) participants use manual approaches to identify functional requirements from the document provided by user. This is due to unavailability of automatic requirement classification tools. Therefore, business analyst is responsible for collecting and identify the functional requirement manually. It is sometimes difficult and risky task because the base of the system depends upon functional requirements.

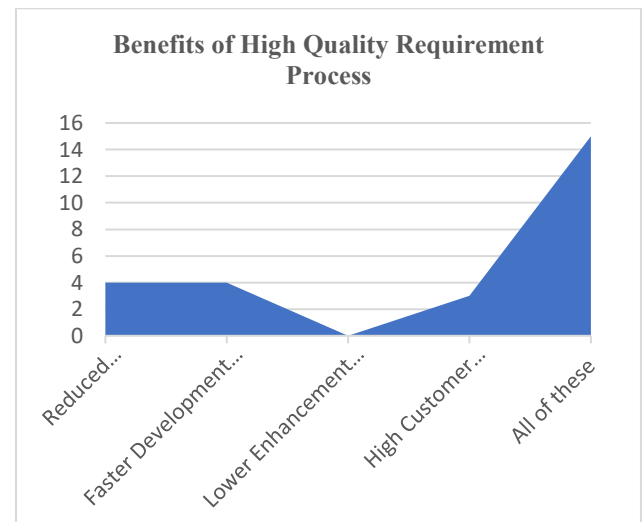


Figure. 2: Benefits of high-quality requirement process

In SQ_5 , we inquired from the participants on how they collect non-functional requirements. The non-functional requirements analyze how well the functional requirements work. Some of the common examples of non-functional requirements are security, performance, reliability, availability etc. The results show that (21/21, 100%) participants utilize manual approaches to identify non-functional requirements from user requirement document. This is true due to unavailability of automatic requirement gathering and classification tool. Therefore, business

analyst is responsible for gathering user requirements and carefully detecting the non-functional requirement.

Moreover, in SQ_6 , we asked the participants about how they gathered ambiguous requirements. Ambiguous requirements are those requirements which is unclear, incomplete, or difficult to understand. According to responses, all of (21/21, 100%) participants said that they use manual approaches to identify ambiguous requirements from user requirement document. This may be due to not presence of automatic requirement gathering and classification tool, business analyst identify the ambiguous requirement manually.

RQ₃ - What are the common risks and challenges faced by Business Analysts during requirements gathering?

We devised survey questions (SQ_{14} , SQ_{15} , SQ_{21} , SQ_{22} , SQ_{23} , SQ_{25}) to answer this research question (RQ_3) as shown in Table I. Further, we examined the responses of these survey questions.

Initially, we wanted to know about the problems arise during requirements gathering (SQ_{22}). According to results obtained, (17/21, 81%) participants faced contradicting or conflicting requirements as main problem followed by undocumented process (12/21, 57.1%) during requirements elicitation. One of the way to resolve conflicting requirements is to ask from the funding sponsor of the project to take decision and resolve the conflict. Further, as shown in figure 3, (10/21, 46.7%) participants encounter communication problems, and (5/21, 23%) participants choose bad requirements as key problems arise in requirement gathering process. A few numbers of participants (3/21, 14%) agreed on *all of these* problems (such as contradicting or conflicting requirements, undocumented process, bad requirements) during gathering requirements.

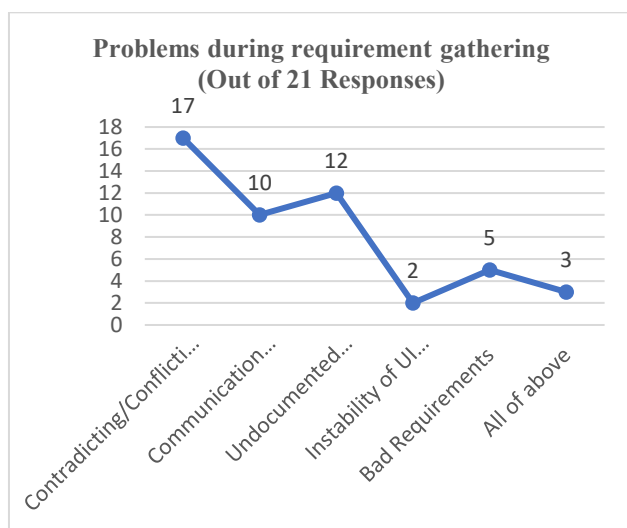


Figure. 3: Challenges during requirement gathering

In SQ_{14} of the survey, we asked the participants whether they perform any risk analysis in requirement gathering

process. Many participants (20/21, 95.2%) agreed that they perform risk analysis. This is because it is very important to measure risk factor on requirements before developing the project.

Finally, in SQ_{15} of the survey, we inquired the participants about risk factor involved in requirement gathering process. As shown in the figure 4, (9/21, 42.9%) participants selected insufficient user involvement and ambiguous requirements as major risks involved in requirement elicitation. This is true in a sense because many projects are failed or remained uncomplete due to insufficient user involvement and ambiguous requirements. Hence the probability of occurring these types of risks always remain high in requirement elicitation process. It is interesting to note that many participants think there are more than one risks that are occurred in requirement elicitation such as insufficient user involvement, inaccurate planning, ambiguous requirement, overlooked stakeholders, and gold plating. Other than insufficient user involvement and ambiguous requirement, (6/21, 28.6%) participants choose inaccurate planning, (2/21, 28.6%) participants agree overlooked stakeholders, and (1/21, 4.8%) participants select gold plating as potential risks during requirement elicitation. The gold plating risk is lower in number because developer generally does not add any new functionality without permission of user.

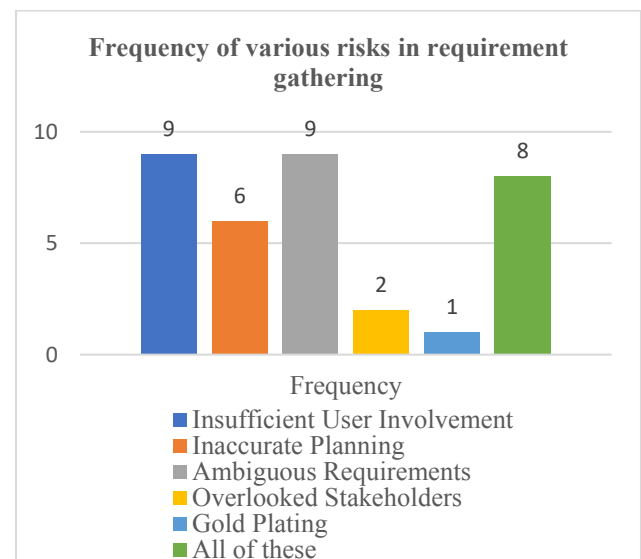


Figure. 4: Risks during requirement gathering

VII. Discussion

This section is dedicated on the discussions of results obtained from our study. As we already know requirement gathering process constitute a major part in software development and well-planned requirement gathering process save more effort, cost, and time. Therefore, it is important to select appropriate requirement gathering approach at right time for project success. For this purpose, we conducted a survey-based study to analyze various requirement gathering aspects. To discuss the major

benefits of our study, we divide the discussion section into two sub sections.

Benefits for Business Analyst: As we conducted this survey from experienced persons of the software industry, so this survey can be helpful of mid-level business analyst and novice to tick all the checks while requirement gathering process. These checks may include what type of best approaches to use for requirement elicitation, what are the problems that may arise, what are the potential risks may come during requirement gathering process, and benefits of high-quality requirement process. Hence, this survey study can be useful for business analyst to choose right techniques and methodologies for requirement gathering.

Moreover, in requirement gathering process, business analyst can use more than one approaches to gather the requirements more efficiently. The business analyst can avoid the potential insufficient user involvement risk by engaging customer more frequently in the requirement elicitation process.

The (15/21, 71.4%) participants agreed that the key benefits of practicing best requirements gathering approaches are higher customer satisfaction, lower enhancement cost, fewer unnecessary and unused features, faster development and reduce development rework.

Benefits for Researcher: This survey study is also helpful for researcher's community because this study highlighted the main aspects of requirement gathering process such as best practices, risk factors, problem occurred, techniques, and methodologies. When a researcher wants to conduct research in requirement elicitation area then this survey study can be helpful or works as a starting point for him to perform further research in this domain such as identify best practices, and risks encounter by business analyst during requirement gathering process.

VIII. Conclusion and Future Work

Requirement gathering is the core and most critical part of software development lifecycle. Well planned requirement gathering process leads to save effort, cost, and time. In this paper, we performed a survey-based study consists of 21 participants to highlight various characteristics of requirement gathering process such as to identify different types of methodologies or approaches business analyst prefer to adopt, how business analyst classify requirements into various types (functional, non-functional, and ambiguous requirements etc.), and potential risks that business analyst may face during requirement elicitation. The main findings showed that interview approach is frequently used by participants (16/21, 76.2%) during requirement gathering process. Further, all (21/21, 100%) participants agreed that they use manual approach to identify functional, non-functional and ambiguous requirements. A lot of participants (9/21, 42.9%). choose insufficient user involvement as the

potential risk faced by business analyst during requirement elicitation. Finally, (15/21, 71.4%) participants agreed that there are various benefits of practicing best approaches such as higher customer satisfaction, lower enhancement cost, fewer unnecessary and unused features, faster development and reduce development rework. To conclude, the selection of right requirement gathering approach plays vital role because the success of any project really on requirement gathering process.

The future work of this research will be focused on performing more surveys on other processes of requirement engineering phases such as requirement specification, requirement validation, and requirement management to analyze different types of practices followed by industry peoples in different requirement phases. Further future research needed to analyze the best combinations of various approaches to elicit the requirements more efficiently during requirement gathering process.

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